

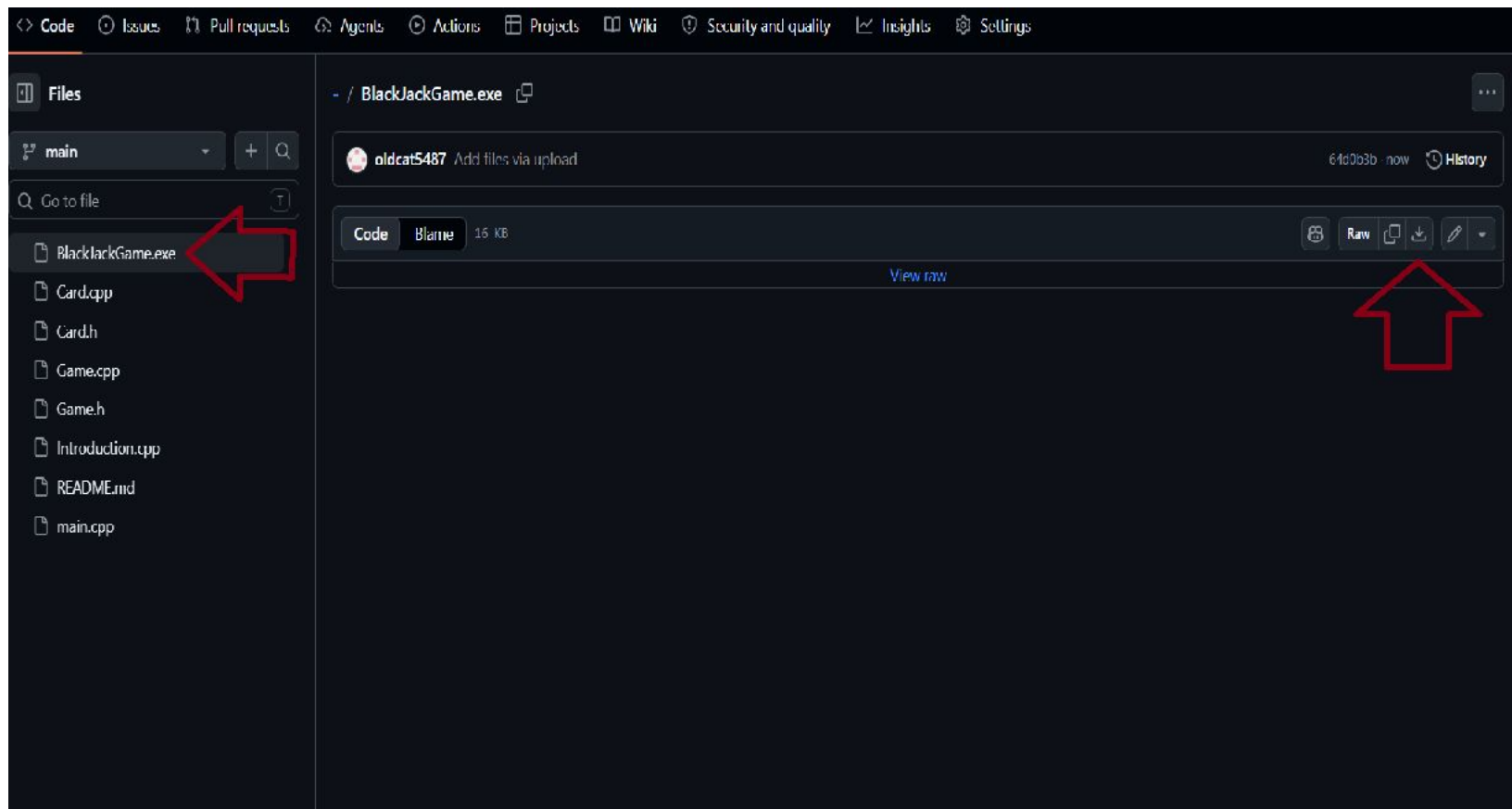
21點

第19組

系級:資工1B

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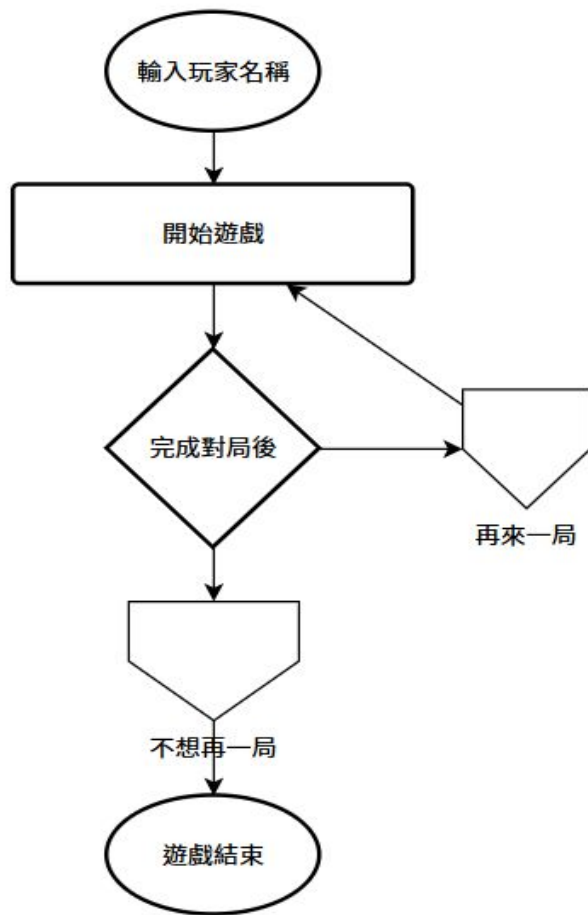
下載方式:



遊戲規則:

此遊戲叫21點,開局雙方都會抽6張牌,目標是讓我方的手牌到21或跟對方比誰最接近21,開局雙方會抽6張牌雙方都棄權後開牌;如果雙方與21相差的點數相同則視為平局,須重開一局,三戰兩勝

流程圖



遊戲遊玩畫面:

```
Welcome to BlackJack
Opponent plays a card
3
The enemy played :3
Current total points of the enemy:3
Your trun
Your choice is:
```

```
1. 5 2.1 3.12 4.12 5.1 6.12 8.Give up
```

```
|
```

遊戲遊玩畫面:

```
1. 5 2.1 3.12 4.12 5.1 6.128.Forfeit
8
Result:
Total enemies:15
Our total:37
The enemy wins
Current enemy wins: 1
Opponent plays a card
3
The enemy played :3
Current total points of the enemy:3
Your trun
Your choice is:
```

```
1. 5 2.1 3.12 4.12 5.1 6.12 8.Give up
1
You chose to play: 5
Our current points:5
8
The enemy played :8
Current total points of the enemy:11
Your trun
Your choice is:
```

```
1. 5 2.1 3.12 4.12 5.1 6.12 8.Give up
|
```

核心程式:

```
7
8 void play_game() {
9     int enemy_win = 0;
10    int player_win = 0;
11
12    int status = 0, enemy_status = 1, player_status = 1;
13
14    int enemy[8] = {};
15    int player[8] = {};
16
17    while (enemy_win < 2 && player_win < 2) {
18
19        /*初始手牌*/
20        enemy[1] = drawCard();
21        enemy[2] = drawCard();
22        enemy[3] = drawCard();
23        enemy[4] = drawCard();
24        enemy[5] = drawCard();
25
26        while (enemy[5] == enemy[1])
27            enemy[5] = drawCard();
28
29        enemy[6] = drawCard();
30
31        while (enemy[6] == enemy[2] || enemy[6] == enemy[1])
32            enemy[6] = drawCard();
33
34        player[1] = drawCard();
35        player[2] = drawCard();
36        player[3] = drawCard();
37        player[4] = drawCard();
38        player[5] = drawCard();
39
40        while (player[5] == player[1])
41            player[5] = drawCard();
42    }
```

核心程式:

```
40 while (player[5] == player[1])
41     player[5] = drawCard();
42
43 player[6] = drawCard();
44
45 while (player[6] == player[2] || player[6] == player[1])
46     player[6] = drawCard();
47 /*確認初始手牌*/
48
49 while (enemy_win < 2 && player_win < 2) {
50     int player_t[8] = {}; //預防會有重複的牌打出
51     int enemy_total = 0;
52     int player_total = 0;
53     int status = 0; //狀態(選擇動作)
54     cout << "Opponent plays a card" << endl;
55     for (int i = 1; i <= 6; i++) { //以敵方為主,最多只會有6個動作
56         if (enemy_total + enemy[i] <= 21) {
57             enemy_status = 1; //檢測敵方狀態(1出牌,0棄權)
58             cout << enemy[i] << endl;
59             enemy_total = enemy_total + enemy[i];
60             cout << "The enemy played :" << enemy[i] << endl;
61             cout << "Current total points of the enemy:" << enemy_total << endl;
62             cout << "Your trun" << endl;
63             cout << "Your choice is:" << endl << endl << endl << endl << endl;
64             cout << "1. " << player[1] << " 2." << player[2] << " 3." << player[3] << " 4." << player[4] << " 5." << player[5] << " 6." << player[6] << " 8.Give
65             cin >> status; //檢測玩家狀態
66
67             if (status == 8) {
68                 player_status = 0;
69                 break;
70             }
71             while (status == 7 || (status == 1 && status <= 6 && player[status] == player_t[status])) {
72                 cout << "You've already typed it" << endl; //你已經出過
73                 cin >> status;
74             }
```


核心程式:

```
73     cin >> status;
74 }
75 cout << "You chose to play: " << player[status] << endl;
76 player_t[status] = player[status];
77 player_total = player_total + player[status];
78 cout << "Our current points:" << player_total << endl;
79 }
80 else {
81     cout << "The other party chose to forfeit" << endl;
82     enemy_status = 0;
83     cout << "Current total points of the enemy:" << enemy_total << endl;
84     cout << "Your trun" << endl;
85     cout << "Your choice is:" << endl << endl << endl << endl;
86     cout << "1. " << player[1] << " 2." << player[2] << " 3." << player[3] << " 4." << player[4] << " 5." << player[5] << " 6." << player[6] << "8.Forfe
87     cin >> status;           //檢測玩家狀態
88
89     if (status == 8) {
90         player_status = 0;
91         break;
92     }
93     while (status == 7 || (status == 1 && status <= 6 && player[status] == player_t[status])) {
94         cout << "You've already played it" << endl;
95         cin >> status;
96     }
97     cout << "You chose to play: " << player[status] << endl;
98     player_t[status] = player[status];
99     player_total = player_total + player[status];
100    cout << "Our current points:" << player_total << endl;
101 }
102 }
103 cout << "Result:" << endl;
104 cout << "Total enemies:" << enemy_total << endl;
105 cout << "Our total:" << player_total << endl;
106 if (21 - enemy_total > 21 - player_total) {
107     enemy_win = enemy_win + 1;
```

核心程式:

```
88
89         if (status == 8) {
90             player_status = 0;
91             break;
92         }
93         while (status == 7 || (status == 1 && status <= 6 && player[status] == player_t[status])) {
94             cout << "You've already played it" << endl;
95             cin >> status;
96         }
97         cout << "You chose to play: " << player[status] << endl;
98         player_t[status] = player[status];
99         player_total = player_total + player[status];
100        cout << "Our current points:" << player_total << endl;
101    }
102 }
103 cout << "Result:" << endl;
104 cout << "Total enemies:" << enemy_total << endl;
105 cout << "Our total:" << player_total << endl;
106 if (21 - enemy_total > 21 - player_total) {
107     enemy_win = enemy_win + 1;
108     cout << "The enemy wins" << endl << "Current enemy wins: " << enemy_win << endl;
109 }
110 else if (21 - enemy_total < 21 - player_total) {
111     player_win = player_win + 1;
112     cout << "We won" << endl << "Current enemy wins: " << player_win << endl;
113 }
114 else cout << "Draw, once again" << endl;
115 }
116 if (enemy_win == 2) cout << "You Lose" << endl;
117 else cout << "You Win" << endl;
118 }
119 cout << "Game Over" << endl;    //結束
120 }
```

工作分類

王晨輝:Game.cpp

武紅英:Card.h

黃馨民:Card.cpp